A PROOF FOR THEOREM 1

A.1 Aided Notations

Prior to the proof, we introduce some notations to assist our illustrations. Define $\hat{g}^t = [\hat{g}_0^t, \hat{g}_1^t, \cdots, \hat{g}_K^t]$ with $\hat{g}_k^t \triangleq \tilde{\nabla}_{\theta_k} l(\Theta^t)$ where $l(\Theta^t) = \frac{1}{B}\sum_{\boldsymbol{x}^i \in \mathcal{S}^t} l(\theta_0, \boldsymbol{h}_1^t, \cdots, \boldsymbol{h}_K^t, y^i)$. One may find that $\hat{g}_k^t$ will not appear during the training since it represents the stochastic gradient based on timely models $\Theta^t = [\theta_0^t, \theta_1^t, \cdots, \theta_K^t]$, which is mainly used as the intermediate result for analytical purpose. Similarly, we denote $\nabla l(\Theta^t) = [\nabla_{\theta_0} l(\Theta^t), \nabla_{\theta_1} l(\Theta^t), \cdots, \nabla_{\theta_K} l(\Theta^t)]$ as the gradients w.r.t. model parameters Θ^t, and $\tilde{g}_k^t = \tilde{\nabla}_{\theta_k} l(\Theta^{t-\tau^t})$ with $\tilde{g}^t = [\tilde{g}_0^t, \tilde{g}_1^t, \cdots, \tilde{g}_K^t]$ as the stochastic gradients based on the stale model parameters.

A.2 Main Proof

Considering that the gradient is L-smooth, we attain:

$$
\begin{aligned}
&\mathbb{E}[l(\Theta^{t+1}) - l(\Theta^t)] \\
&\leq \mathbb{E}[\langle \nabla l(\Theta^t), \Theta^{t+1} - \Theta^t \rangle] + \frac{L'}{2}\mathbb{E}[\|\Theta^{t+1} - \Theta^t\|^2] \\
&= \mathbb{E}\Big[\Big\langle \nabla l(\Theta^t), -\eta \sum_{j=0}^{Q-1} g^{t,j} \Big\rangle\Big] + \frac{L'}{2}\mathbb{E}\Big[\Big\|-\eta \sum_{j=0}^{Q-1} g^{t,j}\Big\|^2\Big] \\
&= -\eta \sum_{j=0}^{Q-1} \mathbb{E}[\langle \nabla l(\Theta^t), g^{t,j} \rangle] + \frac{L'}{2}\eta^2 \mathbb{E}\Big[\Big\|\sum_{j=0}^{Q-1} g^{t,j}\Big\|^2\Big] \\
&\overset{a}{\leq} -\frac{\eta}{2} Q \mathbb{E}[\|\nabla l(\Theta^t)\|^2] - \frac{\eta}{2}\sum_{j=0}^{Q-1} \mathbb{E}[\|g^{t,j}\|^2] \\
&+ \frac{\eta}{2}\sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - g^{t,j}\|^2] + \frac{L'}{2}\eta^2 Q \sum_{j=0}^{Q-1} \mathbb{E}[\|g^{t,j}\|^2] \\
&= -\frac{\eta}{2} Q \mathbb{E}[\|\nabla l(\Theta^t)\|^2] + \frac{\eta}{2}\sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - g^{t,j}\|^2] \\
&- \frac{\eta}{2}(1 - L'\eta Q) \sum_{j=0}^{Q-1} \mathbb{E}[\|g^{t,j}\|^2].
\end{aligned} \tag{17}
$$

where $\overset{a}{\leq}$ comes from the Cauchy–Schwarz inequality. To derive the final result, let us take a closer look at the second term, namely $\frac{\eta}{2}\sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - g^{t,j}\|^2]$.

$$
\begin{aligned}
&\frac{\eta}{2}\sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - g^{t,j}\|^2] = \frac{\eta}{2}\sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - \hat{g}^t + \hat{g}^t - g^{t,j}\|^2] \\
&\leq \eta \sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - \hat{g}^t\|^2] + \eta \sum_{j=0}^{Q-1} \mathbb{E}[\|\hat{g}^t - g^{t,j}\|^2] \\
&\overset{b}{\leq} \eta Q(K+1)\frac{\sigma^2}{B} + \eta \sum_{j=0}^{Q-1} \mathbb{E}[\|\hat{g}^t - g^{t,j}\|^2] \\
&= \eta Q(K+1)\frac{\sigma^2}{B} + \eta \sum_{j=0}^{Q-1} \mathbb{E}[\|\hat{g}_0^t - g_0^{t,j}\|^2] + \eta \sum_{j=0}^{Q-1}\sum_{k=1}^{K} \mathbb{E}[\|\hat{g}_k^t - g_k^{t,j}\|^2] \\
&\leq \eta Q(K+1)\frac{\sigma^2}{B} + \eta L^2 \sum_{j=0}^{Q-1} \mathbb{E}[\|\theta_0^t - \theta_0^{t,j}\|^2] \\
&+ \eta Q L^2 \sum_{k=1}^{K} \mathbb{E}[\|\theta_k^t - \theta_k^{t-\tau_k^t}\|^2] + \eta \sum_{j=0}^{Q-1}\sum_{k=1}^{K} \mathbb{E}[\|\hat{g}_k^t - \tilde{g}_k^t + \tilde{g}_k^t - g_k^{t,j}\|^2] \\
&\leq \eta Q(K+1)\frac{\sigma^2}{B} + \eta L^2 \sum_{j=0}^{Q-1} \mathbb{E}\Big[\Big\|\sum_{n=0}^{j-1} \eta g_0^{t,n}\Big\|^2\Big] \\
&+ \eta Q L^2 \sum_{k=1}^{K} \mathbb{E}\Big[\Big\|\sum_{m=t-\tau_k^t}^{t-1}\sum_{n=0}^{Q-1} -\eta g_k^{m,n}\Big\|^2\Big] \\
&+ 2\eta \sum_{j=0}^{Q-1}\sum_{k=1}^{K} \mathbb{E}[\|\hat{g}_k^t - \tilde{g}_k^t\|^2] + 2\eta \sum_{j=0}^{Q-1}\sum_{k=1}^{K} \mathbb{E}[\|\tilde{g}_k^t - g_k^{t,j}\|^2].
\end{aligned} \tag{18}
$$

One the other hand, we know

$$
\begin{aligned}
&2\eta \sum_{j=0}^{Q-1}\sum_{k=1}^{K} \mathbb{E}[\|\hat{g}_k^t - \tilde{g}_k^t\|^2] \leq 2\eta L^2 K \sum_{j=0}^{Q-1}\sum_{k=1}^{K} \mathbb{E}\Big[\Big\|\theta_k^t - \theta_k^{t-\tau_k^t}\Big\|^2\Big] \\
&= 2\eta Q L^2 K \sum_{k=1}^{K} \mathbb{E}\Big[\Big\|\sum_{m=t-\tau_k^t}^{t-1}\sum_{n=0}^{Q-1} -\eta g_k^{m,n}\Big\|^2\Big] \\
&\overset{c}{\leq} 2\eta^3 Q^2 D L^2 K \sum_{k=1}^{K}\sum_{m=t-\tau_k^t}^{t-1}\sum_{n=0}^{Q-1} \mathbb{E}[\|g_k^{m,n}\|^2] \\
&\overset{d}{\leq} 2\eta^3 Q^3 D^2 L^2 K \sum_{k=1}^{K} \xi_k^2,
\end{aligned} \tag{19}
$$

where the last $\overset{c}{\leq}$ is due to staleness $\tau_k^t \leq D$ and Cauchy–Schwarz inequality, and $\overset{d}{\leq}$ is from the bounded moment in Assumption 2.

We next characterize $2\eta \sum_{j=0}^{Q-1} \sum_{k=1}^{K} \mathbb{E}[\|\tilde{g}_k^t - g_k^{t,j}\|^2]$ as follows.

$$
\begin{aligned}
&2\eta \sum_{j=0}^{Q-1} \sum_{k=1}^{K} \mathbb{E}[\|\tilde{g}_k^t - g_k^{t,j}\|^2] \\
&\leq 2\eta \sum_{j=0}^{Q-1} \sum_{k=1}^{K} \zeta_k^2 \mathbb{E}\big[\big\|\tilde{\nabla}_{\theta_k} \boldsymbol{h}_k(\theta_k^{t-\tau_k^t}) - \tilde{\nabla}_{\theta_k} \boldsymbol{h}_k(\theta_k^{t,j})\big\|^2\big] \\
&\leq 2\eta \sum_{j=0}^{Q-1} \sum_{k=1}^{K} F^2 \zeta_k^2 \mathbb{E}\big[\big\|\theta_k^{t-\tau_k^t} - \theta_k^{t,j}\big\|^2\big] \\
&= 2\eta \sum_{j=0}^{Q-1} \sum_{k=1}^{K} F^2 \zeta_k^2 \mathbb{E}\Big[\Big\|-\eta \sum_{m=t-\tau_k^t}^{t-1} \sum_{n=0}^{Q-1} g_k^{m,n} - \eta \sum_{n=0}^{j-1} g_k^{t,n}\Big\|^2\Big] \\
&\leq 2\eta^3 \Big(Q^3 D^2 + Q^3 D + \frac{1}{3} Q^3 - Q^2 D - \frac{1}{2} Q^2 + \frac{1}{6} Q\Big) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2 \\
&\leq \eta^3 Q^3 (2D^2 + 2D + 1) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2.
\end{aligned}
\tag{20}
$$

Similarly, we can obtain the rest terms in Eq. (18), which are $\eta L^2 \sum_{j=0}^{Q-1} \mathbb{E}[\| \sum_{n=0}^{j-1} \eta g_0^{t,n}\|^2] \leq \frac{1}{3}\eta^3 Q^3 L^2 \xi_0^2$ and $\eta Q L^2 \sum_{k=1}^{K} \mathbb{E}[\| \sum_{m=t-\tau_k^t}^{t-1} \sum_{n=0}^{Q-1} -\eta g_k^{m,n}\|^2] \leq \eta^3 Q^3 D^2 L^2 \sum_{k=1}^{K} \xi_k^2$, respectively. Therefore, substituting Eqs. (19) and (20) into Eq. (18) yields

$$
\begin{aligned}
&\frac{\eta}{2} \sum_{j=0}^{Q-1} \mathbb{E}[\|\nabla l(\Theta^t) - g^{t,j}\|^2] \leq \eta Q (K+1) \frac{\sigma^2}{B} + \frac{1}{3}\eta^3 Q^3 L^2 \xi_0^2 \\
&+ \eta^3 Q^3 D^2 L^2 (1+2K) \sum_{k=1}^{K} \xi_k^2 + \eta^3 Q^3 (2D^2 + 2D + 1) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2.
\end{aligned}
\tag{21}
$$

Following the same vein, we substitute Eq. (21) into the main routine of Eq. (17) to obtain:

$$
\begin{aligned}
&\mathbb{E}[l(\Theta^{t+1}) - l(\Theta^t)] \leq -\frac{\eta}{2} Q \mathbb{E}[\|\nabla l(\Theta^t)\|^2] + \eta Q (K+1) \frac{\sigma^2}{B} \\
&+ \frac{1}{3}\eta^3 Q^3 L^2 \xi_0^2 + \eta^3 Q^3 D^2 L^2 (1+2K) \sum_{k=1}^{K} \xi_k^2 \\
&+ \eta^3 Q^3 (2D^2 + 2D + 1) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2 - \frac{\eta}{2}(1 - L'\eta Q) \sum_{j=0}^{Q-1} \mathbb{E}[\|g^{t,j}\|^2] \\
&\overset{e}{\leq} -\frac{\eta}{2} Q \mathbb{E}[\|\nabla l(\Theta^t)\|^2] + \eta Q (K+1) \frac{\sigma^2}{B} + \frac{1}{3}\eta^3 Q^3 L^2 \xi_0^2 \\
&+ \eta^3 Q^3 D^2 L^2 (1+2K) \sum_{k=1}^{K} \xi_k^2 + \eta^3 Q^3 (2D^2 + 2D + 1) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2,
\end{aligned}
\tag{22}
$$

in which $\overset{e}{\leq}$ is because $0 < \eta \leq \frac{1}{L'Q}$. We carry out a telescope sum from $t = 0$ to $t = T - 1$ of both sides and then:

$$
\begin{aligned}
&\mathbb{E}[l(\Theta^T) - l(\Theta^0)] \leq -\frac{\eta Q}{2} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla l(\Theta^t)\|^2] \\
&+ T\eta Q (K+1) \frac{\sigma^2}{B} + \frac{T}{3}\eta^3 Q^3 L^2 \xi_0^2 \\
&+ T\eta^3 Q^3 D^2 L^2 (1+2K) \sum_{k=1}^{K} \xi_k^2 + T\eta^3 Q^3 (2D^2 + 2D + 1) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2.
\end{aligned}
\tag{23}
$$

Rearrange terms and multiply both sides by $\frac{2}{T\eta Q}$ to deduce Eq. (2).

As for Eq. (3), we can just use PL condition in Assumption 4 to replace each $l(\Theta^t) - l(\Theta^*) \leq \frac{\mathbb{E}[\|\nabla l(\Theta^t)\|^2]}{2\mu}$.

B PROOF FOR THEOREM 2

If we can show that the server ends up processing the embedding vectors $\{\boldsymbol{h}_1^t, \cdots, \boldsymbol{h}_K^t\}, \forall t > D$ at $\max_{k\in\mathcal{K}}\{t_k^e + t_k^{ec}\} + tQt^{sc}$, and meanwhile has received the next embedding $\{\boldsymbol{h}_1^{t+1}, \cdots, \boldsymbol{h}_K^{t+1}\}$, then the form of wall-clock time stands. Next, we prove this by induction.

First of all, for $\{\boldsymbol{h}_1^t, \cdots, \boldsymbol{h}_K^t\}, t = D + 1$, the conclusion holds as analyzed before. We will illustrate that when the result still stands for $\{\boldsymbol{h}_1^t, \cdots, \boldsymbol{h}_K^t\}, t > D + 1$, then it is true for $\{\boldsymbol{h}_1^{t+1}, \cdots, \boldsymbol{h}_K^{t+1}\}$. For data samples $\mathcal{S}^{t+1}$, suppose the embedding $\boldsymbol{h}_k^t$ is sent out at time F^t and received by the server at $F^t + t_k^{ec}$. Since processing $\{\boldsymbol{h}_1^t, \cdots, \boldsymbol{h}_K^t\}$ is blocked as PS is busy handling embedding vectors, which implies $F^t + t_k^{ec} < \max_{k\in\mathcal{K}}\{t_k^e + t_k^{ec}\} + tQt^{sc}$. Also, PS will complete model updating at $\max_{k\in\mathcal{K}}\{t_k^e + t_k^{ec}\} + (t+1)Qt^{sc}$.

The only situation where PS receives embedding $\{\boldsymbol{h}_1^{t+1}, \cdots, \boldsymbol{h}_K^{t+1}\}$ while staying idle is that each party needs to first update local model and then learn $\boldsymbol{h}_k^{t+1}$. In this case, $\boldsymbol{h}_k^{t+1}$ will be generated at $F^t + Qt_k^{pc} + t_k^e$ and arrives at PS at $F^t + Qt_k^{pc} + t_k^e + t_k^{ec}$. However, we know that $\max_{k\in\mathcal{K}}\{t_k^e + t_k^{ec}\} + (t+1)Qt^{sc} = \max_{k\in\mathcal{K}}\{t_k^e + t_k^{ec}\} + tQt^{sc} + Qt^{sc} \geq F^t + t_k^{ec} + Qt^{sc} \geq F^t + Qt_k^{pc} + t_k^e + t_k^{ec}$, indicating that the idle situation will not happen. Therefore, $\{\boldsymbol{h}_1^{t+1}, \cdots, \boldsymbol{h}_K^{t+1}\}$ will be firstly buffered and then executed at $\max_{k\in\mathcal{K}}\{t_k^e + t_k^{ec}\} + (t+1)Qt^{sc}$. In other words, the form of wall-clock time is still correct. By induction, we can derive the final result.

C PROOF FOR THEOREM 3

The convergence derivation in the l-stage is similar to that of the stable environment, where the main difference lies in that previous local step Q_{l-1} will influence the l-th stage owing to asynchronous model updating. Following the same idea as obtaining Eq. (22), for

$t \in [T_{l-1}, T_{l-1} + D)$, we attain:

$$
\begin{aligned}
&\mathbb{E}[l(\Theta^{t+1}) - l(\Theta^t)] \\
&\leq -\frac{\eta}{2} Q_l \mathbb{E}[\|\nabla l(\Theta^t)\|^2] + \eta Q_l (K+1) \frac{\sigma^2}{B} + \frac{1}{3} \eta^3 Q_l^3 L^2 \xi_0^2 \\
&+ \eta^3 Q_l [(T_{l-1} + D - t) Q_{l-1} + (t - T_{l-1}) Q_l]^2 L^2 (1 + 2K) \sum_{k=1}^{K} \xi_k^2 \\
&+ 2\eta^3 \sum_{j=0}^{Q_l - 1} [(T_{l-1} + D - t) Q_{l-1} + (t - T_{l-1}) Q_l + j]^2 F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2 \\
&\leq -\frac{\eta}{2} Q_l \mathbb{E}[\|\nabla l(\Theta^t)\|^2] + \eta Q_l (K+1) \frac{\sigma^2}{B} + \frac{1}{3} \eta^3 Q_l^3 L^2 \xi_0^2 \\
&+ \eta^3 Q_l [2(T_{l-1} + D - t)^2 Q_{l-1}^2 + (t - T_{l-1})^2 Q_l^2] L^2 (1 + 2K) \sum_{k=1}^{K} \xi_k^2 \\
&+ 2\eta^3 Q_l [2(T_{l-1} + D - t)^2 Q_{l-1}^2 + 2(t - T_{l-1} + 1)^2 Q_l] F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2.
\end{aligned} \tag{24}
$$

Regarding $t \in [T_{l-1} + D, T_l)$, we analogously obtain:

$$
\begin{aligned}
&\mathbb{E}[l(\Theta^{t+1}) - l(\Theta^t)] \\
&\leq -\frac{\eta}{2} Q_l \mathbb{E}[\|\nabla l(\Theta^t)\|^2] + \eta Q_l (K+1) \frac{\sigma^2}{B} + \frac{1}{3} \eta^3 Q_l^3 L^2 \xi_0^2 \\
&+ \eta^3 Q_l^3 D^2 L^2 (1 + 2K) \sum_{k=1}^{K} \xi_k^2 + \eta^3 Q_l^3 (2D^2 + 2D + 1) F^2 \sum_{k=1}^{K} \zeta_k^2 \xi_k^2.
\end{aligned} \tag{25}
$$

Based on Eqs. (24) and (25), adding both sides from T_{l-1} to $T_l - 1$ yields Eq. (14). Similarly, using PL condition we can derive Eq. (15).